KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

COLLEGE OF HEALTH SCIENCES

SCHOOL OF MEDICAL SCIENCES

DEPARTMENT OF OBSTETRICS AND GYNAECOLOGY

BSC. PHYSICIAN ASSISTANTSHIP

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AWARENESS, RISK PERCEPTION AND PREVENTIVE MEASURES FOR SEXUALLY TRANSMITTED INFECTIONS AMONG KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY STUDENTS

A RESEARCH PROJECT SUBMITTED TO THE DEPARTMENT OF OBSTETRICS AND GYNAECOLOGY, KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR BACHELOR OF SCIENCE (HONS) PHYSICIAN ASSISTANTSHIP DEGREE

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DECLARATION

We hereby declare that this project is the result of our own independent investigation. Except for the works of others, which have been duly acknowledged in the reference section, this project has not been previously submitted for any award. It was completed under the supervision of Dr. B. B. Ofosu Barko and is entirely original.

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CHAPTER ONE

INTRODUCTION

1.1 BACKGROUND OF THE STUDY

Sexually Transmitted Infections, also known as STIs, are infections caused by bacteria, viruses, parasites or fungi that are mainly transmitted through sexual contact, including vaginal, anal and oral sex. Some STIs can be transmitted through non-sexual means, such as from mother to child during childbirth, through blood transfusions or shared needles (World Health Organization, 2025).

According to the World Health Organization (WHO), eight pathogens are responsible for the highest rates of STIs. Among them, four bacterial or parasitic infections—syphilis, chlamydia, gonorrhoea and trichomoniasis—are currently curable. The remaining four—hepatitis B, herpes simplex virus, HIV and human papilloma virus—are viral infections that cannot be cured but can be managed (World Health Organization, 2025).

Sexually transmitted infections (STIs) are a significant public health concern around the world, infecting millions of individuals each year. The World Health Organization (WHO) states that over 1 million new sexually transmitted infection cases are reported daily, with young people between 15–49 years being the most infected (World Health Organization, 2025).

STIs such as syphilis, gonorrhoea, chlamydia, and Human Papilloma Virus (HPV) can result in severe health consequences, including infertility, chronic pain, and an increased risk of HIV acquisition (Newman et al., 2015) .The prevalence of delayed treatment of STIs in sub-Saharan Africa was 47% in 2024, with factors such as poor knowledge about STIs, low education levels, and misconceptions about STI causes contributing to this delay; these factors highlight the significant role of ignorance and limited access to health services in the persistence of STIs in the region (Agimas et al., 2024).In Ghana, approximately 3.4% of the population is affected by sexually transmitted infections (STIs). Among individuals exhibiting symptoms, prevalence rates can be as high as 28%. According to the 2022 Ghana Demographic and Health Survey, the prevalence of STIs among adolescents aged 15–19 has increased over time: the prevalence in females rose from 2.2% in 2003 to 7.1% in 2022, and in males from 2.2% in 2003 to 12.5% in 2022 (Ghana Statistical Service, 2022).

Sexually active men aged 25–34 years, those whose age at first sexual intercourse was below 20 years, and those with two or more sexual partners apart from their spouse had higher odds of reporting STIs (Seidu et al., 2021). University students, particularly those in Kwame Nkrumah University of Science and Technology (KNUST), might be at a higher risk of STI contraction because research on adolescents in Ghana highlighted that peer influences significantly affect sexual activityThe study found that peer pressure was a notable factor influencing sexual behaviour among adolescents, which could extend to university students as well (Bingenheimer, Asante and Ahiadeke, 2015). It is therefore necessary to establish the awareness, risk perceptions, and prevention practices of students in order to formulate effective sexual health interventions based on their needs.

1.2 PROBLEM STATEMENT

The rising incidence of STIs among young adults in Ghana justifies the examination of awareness and preventive behaviour within this population (Ghana Statistical Service, 2022). Although research on STI knowledge and risk perception among African youths has been conducted to some extent, not many have focused on university students in Ghana, and particularly at KNUST. Low awareness and common misconceptions about STIs can lead to risky sexual activities, which in turn accelerates transmission levels and the danger of severe health issues (Amo Adjei & Tuoyire, 2016).

Moreover, while some students have general knowledge of STIs, their own risk perception is often low, and this may result in a lack of preventive measures. This discrepancy between knowledge and behaviour raises concern about the quality of existing sexual health education programs and the access to STI prevention services for college students. Despite high levels of awareness about HIV/AIDS among university students, this knowledge does not consistently translate into preventive behaviours. A study involving 324 students at a tertiary institution in Accra revealed that while participants demonstrated a mean score of 7.7 out of 12 on HIV/AIDS knowledge questions, many lacked understanding about the causative agent of AIDS. Notably, 45% of students had never undergone HIV testing, indicating a gap between knowledge and action (Asante and Oti-Boadi, 2013).This research aims to assess STI awareness, risk perception, and preventive behaviour among KNUST students to aid in the formulation of more efficient health education strategies and interventions.

1.3 RESEARCH OBJECTIVES

This research aims to:

1. Investigate the level of awareness about STIs among KNUST students.

2. Assess students' perceptions of the risks of STIs.

3. Assess preventive behaviours students undertake to avoid STI transmission.

4. Identify main factors influencing students' knowledge, risk perception, and preventive behaviours.

1.4 RESEARCH QUESTIONS

1. What is the level of awareness of STIs among KNUST students?

2. How do students perceive their risk of contracting STIs?

3. What are the preventive behaviours students use to curtail the risk of STI transmission?

4. What are the determinants of students' awareness, risk perception, and preventive behaviours?

1.5 SIGNIFICANCE OF THE STUDY

The study is significant in the following ways:

- It provides data on the level of STI awareness and risk perception among university students, contributing to strengthened health education programs.

- Findings from this research can help university managers, policy makers, and health practitioners formulate targeted interventions promoting safer sexual behaviours.

- The study adds to the body of literature on STI prevention efforts among young adults in Ghana, plugging existing knowledge gaps.

- The results can inform efforts to improve access to STI-related health care services, such as screening, treatment, and counselling of university students.

1.6 SCOPE OF THE STUDY

The study will focus on undergraduate and postgraduate students of KNUST, their knowledge, perceptions, and behaviour towards STIs.

Surveys and interviews will be used to collect data from students of various academic disciplines and levels of study. Even though the study will focus mainly on STIs, it will not address broader issues of reproductive health. The study will primarily examine:

- Knowledge, attitudes, and practices regarding STIs.
- Risk factors (e.g., unprotected sex, multiple partners).
- Preventive behaviors (e.g., condom use, screening).
- Access to and use of STI-related services.

1.7 LIMITATIONS

1. This study is only limited to students of KNUST so it potentially limits the generality of findings.

2. Participants may be hesitant to disclose certain behaviors or experiences due to social desirability bias and sensitivity of the topic.

3. Time constraints can influence the depth and comprehensiveness of this research.

4. Data collected through self-reports may be subject to recall bias where individuals inaccurately remember past events or experience.

1.8 BASIC ASSUMPTIONS

1. The sample of students used in the study will be a true reflection of the total number of students at KNUST.
2. All answers provided by respondents are assumed to be accurate despite potential self-reporting bias.

CHAPTER TWO

LITERATURE REVIEW

2.1 OVERVIEW OF SEXUALLY TRANSMITTED INFECTIONS (STIs)

Sexually transmitted infections (STIs) pose a significant global health challenge, with the World Health Organization reporting nearly one million new cases each day (WHO, 2019). Common infections including chlamydia, gonorrhoea, syphilis, trichomoniasis, HIV, and HPV differ in severity, and if untreated, some can lead to serious long-term outcomes such as infertility or cancer (CDC, 2023; WHO, 2022). These infections are mainly spread through sexual contact, and having one STI can elevate the likelihood of contracting additional infections, including HIV (CDC, 2023). Preventative measures such as condom use, routine screening, partner notification, and vaccination—particularly against HPV—are key strategies in reducing transmission (Shew et al., 2006).. However, controlling STIs remains difficult due to issues like antibiotic resistance (especially in gonorrhoea), social stigma, and inadequate healthcare infrastructure in low-resource areas (Lancet Commission on STIs, 2021; BMC Infectious Diseases, 2022). Reducing the global impact of STIs requires a comprehensive approach that includes public health education, supportive policy development, and broader access to healthcare services (World Health Organization, 2022).

2.2 GLOBAL BURDEN OF STIs

Sexually transmitted infections (STIs) continue to be a serious global health issue, with over a million new infections occurring every day most from treatable diseases like chlamydia, gonorrhoea, syphilis, and trichomoniasis (World Health Organization [WHO], 2024Viral STIs, such as human papillomavirus (HPV) and herpes simplex virus (HSV), are widespread and contribute to around 2.5 million deaths and 1.2 million cancer cases each year (World Health

Organization, 2024). One of the most pressing challenges is antimicrobial resistance, especially in gonorrhoea, where resistance to the last-line antibiotic ceftriaxone has been reported in up to 40% of cases in some countries (WHO, 2024). Certain groups like men who have sex with men, sex workers, and people who inject drugs are disproportionately affected, often facing stigma and poor access to healthcare (WHO, 2024).

2.3 AWARENESS OF STIs

A study assessing STI knowledge among university students revealed that awareness levels were notably low for several STIs, with only 32.5% correctly identifying that HPV can cause genital warts (Alshemeili et al., 2023). In a study in Ghana, Addo and Tagoe (2020) came up with an idea that although awareness of HIV is more than 90%, only 46% of participants knew of HPV and just 34%were aware of syphilis. Some relevant sources of STI awareness are media campaign, peer education, and school-based programs, even though misinformation continues to exist (Aziato and Ohemeng,2020). A study across several countries found that awareness of gonorrhea was high among both men (96%) and women (85%). Despite this high awareness, the actual prevalence of gonorrhea was low, suggesting that knowledge alone may not be sufficient to reduce infection rates (Vasudeva et al., 2022).

2.4 KNOWLEDGE OF STIs

Knowledge includes awareness of transmission, symptoms, repercussions, and avoidance of STIs. Studies have shown significant gaps in STI knowledge, even among literates. Ezeanolue et al.

(2015) found that Nigerian adolescents regard STIs could be transmitted through mosquito bites, kissing, or sharing food. Likewise, Tenkorang (2014) noticed that misbeliefs were common among Ghanaian youth. Competent knowledge is certainly associated with systematic education, sexual health instruction, and open communication with medical professionals.

2.5 RISK PERCEPTION OF STIs

Risk perception refers to how individuals assess their likelihood of contracting sexually transmitted infections and the severity of the associated consequences. It plays a critical role in shaping sexual behaviour and decision-making. Studies among Ghanaian adolescents have shown that low perceived risk often leads to engagement in unsafe sexual practices, despite general awareness of STI-related dangers (Barimah et al., 2022).

2.6 IMPLICATIONS FOR PUBLIC HEALTH

Gaining insight into individuals' levels of awareness, knowledge, and risk perception regarding sexually transmitted infections (STIs) is critical for developing and implementing effective public health interventions. Successful STI prevention and control require a comprehensive, multi-layered approach that not only educates the public but also encourages sustainable behavioural change. This includes the integration of targeted educational programs, mass media campaigns, peer-led initiatives, and easily accessible sexual health services (World Health Organization, 2022).

Rashid and Rani (2021) emphasize the importance of culturally relevant and age-appropriate communication strategies, which can be delivered through schools, community outreach, digital platforms, and social media to maximize reach and impact. These strategies should be tailored to resonate with specific populations, taking into account cultural norms, language, and local beliefs that may influence sexual behaviour.

Incorporating theoretical frameworks such as the Health Belief Model (HBM) and the Theory of Planned Behaviour (TPB) (Ajzen and Fishbein, 1980) can further enhance the effectiveness of interventions. These models help explain how individual beliefs, perceived risks, social norms, and self-efficacy influence health-related behaviours. For instance, the HBM highlights the importance of perceived severity and susceptibility in motivating behaviour change, while the TPB emphasizes the role of attitudes, perceived behavioural control, and intentions.

Moreover, public health programs should focus on reducing stigma surrounding STIs, promoting routine testing, and ensuring that youth and vulnerable populations have confidential access to treatment and counselling. Collaboration between health institutions, educational bodies, and community organizations is essential to create supportive environments that empower individuals to make informed sexual health choices. Overall, a strategic and evidence-based approach is key to reducing STI transmission and improving sexual health outcomes on a broader scale

2.7 CONCLUSION

In summary, sexually transmitted infections (STIs) remain an urgent and growing global health crisis, fuelled by alarmingly high infection rates, limited awareness, and the mounting threat of antimicrobial resistance. Although for instance HIV awareness is relatively high, far too many people especially young individuals lack basic knowledge about STIs like HIV and how they spread. Misguided perceptions of risk, often shaped by culture, emotions, and social norms, continue to weaken the impact of prevention efforts, even in the face of well-intentioned education campaigns. Addressing these gaps through culturally sensitive, evidence-based public health strategies is crucial to reducing STI transmission and improving global sexual health outcomes.

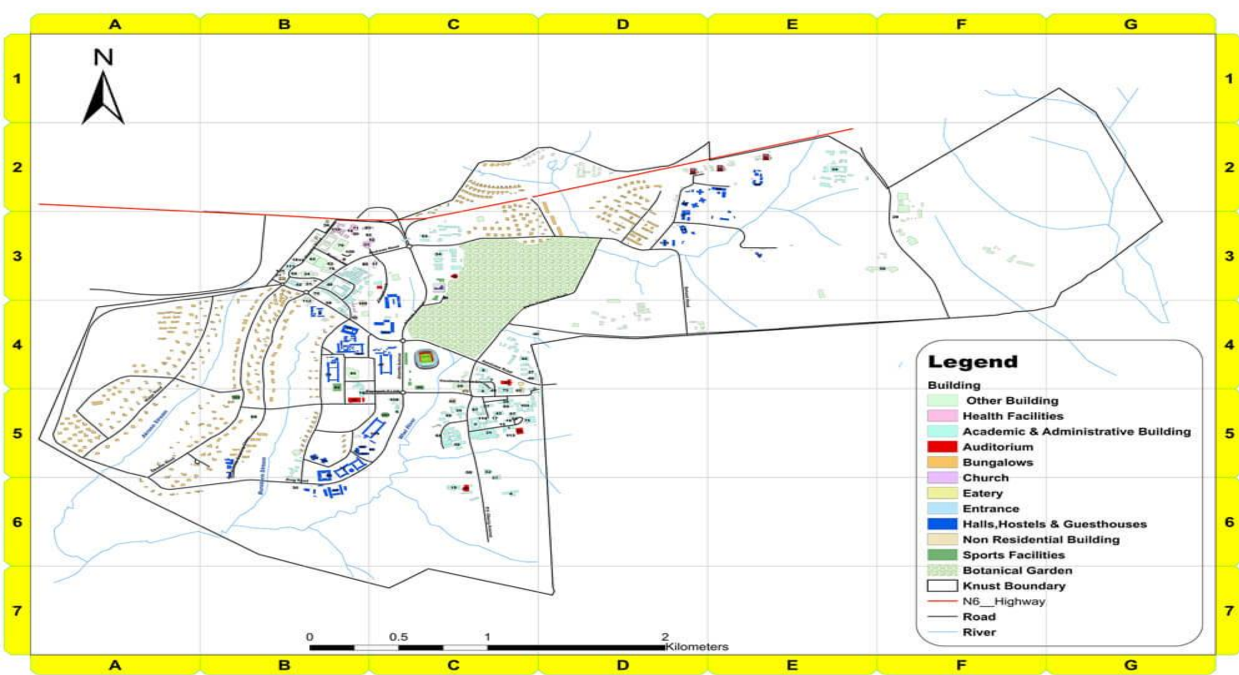
CHAPTER THREE

PROFILE OF STUDY AREA

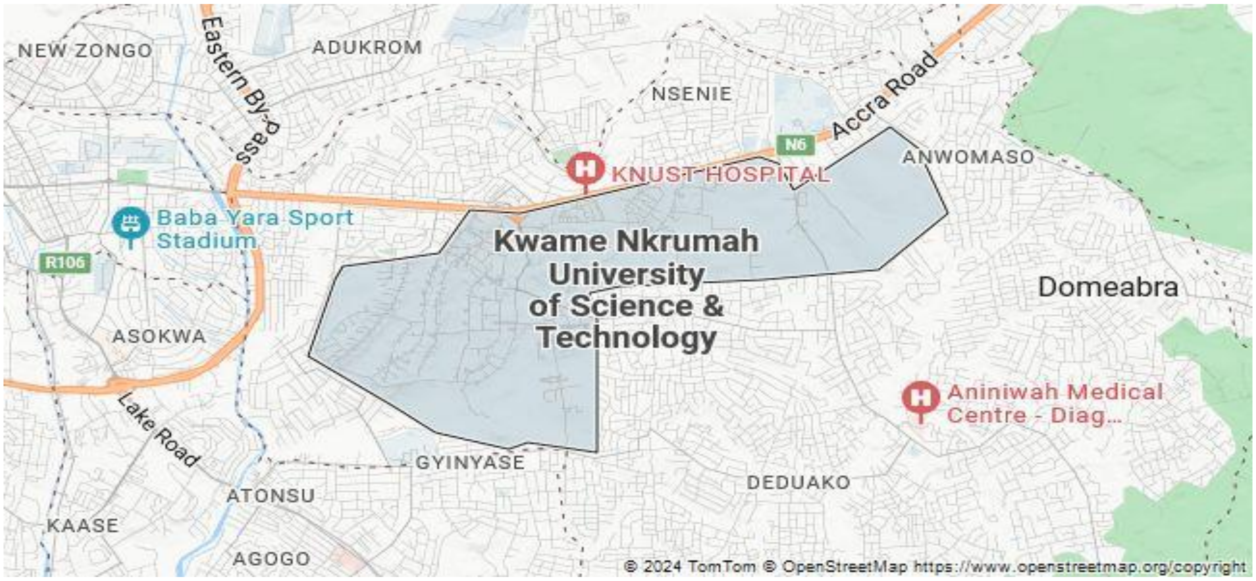
3.1 LOCATION

Kwame Nkrumah University of Science and Technology (KNUST), a leading public institution in West Africa celebrated for its excellence in science and technology education, provided an ideal setting for this study. The university's rich diversity, spanning various academic disciplines, regions, and ethnicities offer valuable insights into how such factors can shape health-related behaviours, making KNUST a compelling and relevant choice for this research.

Kwame Nkrumah University of Science and Technology (KNUST), established in 1952, is located approximately eight kilometres from the centre of Kumasi, the capital of the Ashanti Region. The university occupies a total land area of 2,512.96 acres (1,016.96 hectares), equivalent to approximately eighteen square kilometres. It is bordered by several neighbouring towns, including Ayeduase, Kotei, Kentinkrono, Ayigya, Boadi, and Ahinsan, which contribute to its strategic location and accessibility within the region.



Map1: Showing KNUST Campus



Map 2: showing the location of KNUST in Kumasi

3.2 POPULATION

According to statistics from QAPO KNUST in 2022, the university has a student population of approximately 89,486 which includes postgraduates and undergraduates with 35,514 (39.69%) representing the female population and 53,972 (60.31%) representing the male population.

3.3 ACCOMMODATION AND HOUSING

The Kumasi campus of KNUST comprises six traditional halls of residence, each administered by a hall council made up of both senior and junior members of the university community. These halls include Unity Hall (Conti), Independence Hall, Republic Hall, Africa Hall, Queens Hall, and University Hall (Katanga). Each hall is overseen by a Hall Master, supported by a Senior Tutor, Hall Bursar, and other administrative personnel.

In addition to the traditional halls, the campus hosts several university-affiliated hostels such as the GUSSS Hostels (Brunei and Chancellor's Hall), the Mastercard Foundation Hostel (IMPACT Building), the SRC Hostel, and the Tek Credit Hostel. Furthermore, a number of university-approved private hostels situated in close proximity to the main campus offer accommodation to students who are not assigned places in the on-campus halls of residence.

3.4 HEALTHCARE

Kwame Nkrumah University of Science and Technology (KNUST) offers comprehensive healthcare services to its student body through both the KNUST Hospital and the on-campus Student Clinic. The KNUST Hospital, conveniently located on the university campus, provides a full range of medical services, including outpatient and inpatient care, emergency response, and specialist consultations in areas such as general medicine, dentistry, gynaecology, and ophthalmology. In support of student mental health, the facility also offers counselling and psychological services aimed at fostering emotional well-being. Meanwhile, the KNUST Student Clinic delivers accessible primary healthcare—such as routine check-ups, immunizations, treatment of minor ailments, and health education—ensuring that students have convenient, timely care for everyday concerns without needing to visit the main hospital.

To further promote student health, the university organizes regular health education initiatives and periodic medical screenings. In addition, on-campus pharmacies are available to ensure timely access to prescribed medications. Medical services are accessible 24 hours a day, seven days a week, for both students and academic staff, guaranteeing continuous and reliable care.

CHAPTER FOUR

METHODOLOGY

4.1 STUDY DESIGN

For this study, a descriptive cross-sectional approach was chosen. This is due to its very high time efficacy. It also records information at a certain moment in time. Lastly, previous researchers have used this study design for this topic or related topics and it has shown to be successful.

4.2 STUDY POPULATION

The target population comprised all students at KNUST, including undergraduates and postgraduates.

4.2.1 INCLUSION CRITERIA

1. It included currently registered students at Kwame Nkrumah University of Science and Technology (KNUST).

2. It included participants between 18 and 45 years old to ensure legal capacity to provide informed consent.

3. It included participants who were willing and able to provide written informed consent.

4. It included all genders with no restriction based on sexual-activity status.

5. It included only participants who voluntarily agreed to complete the study questionnaire or interview.

4.2.2 EXCLUSION CRITERIA

1.It excluded students with mental illness to ensure that all participants could provide valid informed consent and maintain data integrity.

2. Non-Students: Individuals who are not currently in any undergraduate or postgraduate program in KNUST (e.g., alumni, staff, or external participants).

4.3 SAMPLE SIZE

A sample size of 332 students of KNUST were recruited for this study.

The sample size was determined using the Charan and Biswas' formula to calculate a representative sample size,

which is; $n= Z^2(p)(1-p)/ E^2$

Where n = Sample size,

Z = z value corresponding to a 95% level of significance = 1.96

p = expected proportion of population = 0.50

$(1 – p) = 0.50$

E= absolute precision (5% = 0.05)

As a result, the sample size from the above is $n = (1.96)^2 \times 0.50 \times 0.50 / 0.05^2 = 384.16$ approximately 384 students

Adding 10% non-response rate = 422 students in total.

4.4 SAMPLING METHOD

A simple random sampling was used to select participants from Kwame Nkrumah University of Science and Technology. This sampling techniques was chosen because it is convenience and represents the population.

4.5 DATA COLLECTION TECHNIQUE

A well-structured self-administered questionnaire was the method and tool utilized to collect the data. The questionnaire consisted of four sections A, B, C and D. Section A will contain Demographics (Age, gender, relationship status, religion, year of study, college and program of study). Section B sought to ascertain answers to questions on the general knowledge on Sexually Transmitted Infections (STIs) and Section C tried ascertain answers to questions on their perceptions about STIs. Section D sought to ascertain answers to questions on how their knowledge and perceptions influence their reproductive health behaviours and decisions. Participants were given direct link through WhatsApp and some completed them on our phones.

4.6 DATA ANALYSIS

Data obtained were analysed using Statistics and Data Science (STATA 17.0). Appropriate tabulations, chart and other diagrams were generated using the above-mentioned software to make interpretations and analysis easier.

4.7 ETHICAL CONSIDERATION

Informed consent was obtained from all participants, after ensuring they understood the purpose of the study and that their participation was voluntary. Their privacy was protected as their personal information are kept confidential and secured. The data obtained was used only for the intended purpose and no harm will come to the participants in any way.

CHAPTER FIVE

RESULTS

5.1 INTRODUCTION

This chapter presents the results of the study on awareness, risk perceptions, and preventive practices toward sexually transmitted infections (STIs) among the participants. The data were obtained through questionnaires designed to assess respondents' level of knowledge, their perception of susceptibility and severity of STIs, as well as the preventive measures they adopt. The findings are organized and presented in line with the study objectives, beginning with the demographic characteristics of respondents, followed by their awareness and knowledge of STIs, their perception of associated risks, and finally the preventive practices employed. These results provide a basis for understanding the current situation and will form the foundation for the subsequent discussion and recommendation.

5.2 DEMOGRAPHIC PROFILE OF PARTICIPANTS

The study involved 333 consenting students, with the majority being young adults. The highest concentration of participants was in the 20–22 age range, particularly at ages 21 (59 participants), 22 (64 participants), and 20 (56 participants). Participation sharply declined after age 22, with very few participants above age 25 and minimal representation from ages 30 to 45.

In terms of gender, the distribution was nearly equal: 49.9% females (n=166) and 48.9% males (n=163), while 1.2% (n=4) preferred not to disclose their gender. Most participants were undergraduates (94.9%), with Level 100 (27.6%) and Level 400 (25.8%) students making up the largest groups, while only 5.1% (n=17) were postgraduates.

Participants came from various colleges, with the College of Health Sciences contributing the highest proportion (40.2%), followed by Humanities (29.7%) and Engineering (23.1%). Smaller contributions came from Arts, Built Environment, and Agriculture.

Regarding relationship status, most respondents were single (67.9%), while 30.6% were in a relationship and 1.5% were married. When it comes to sexual activity, 67.6% reported not engaging in sex, while 25.2% were in committed relationships, 4.8% engaged in casual sex, and 2.4% were married. Additionally, 57.4% had no sexual partner in the last two years, while 25.2% had one partner and 9.9% had two. In terms of lifetime sexual history, 52.0% had never had a sexual partner, 20.1% had one partner, and 6.3% had six or more.

Figure 5.2.1 – Age of participants

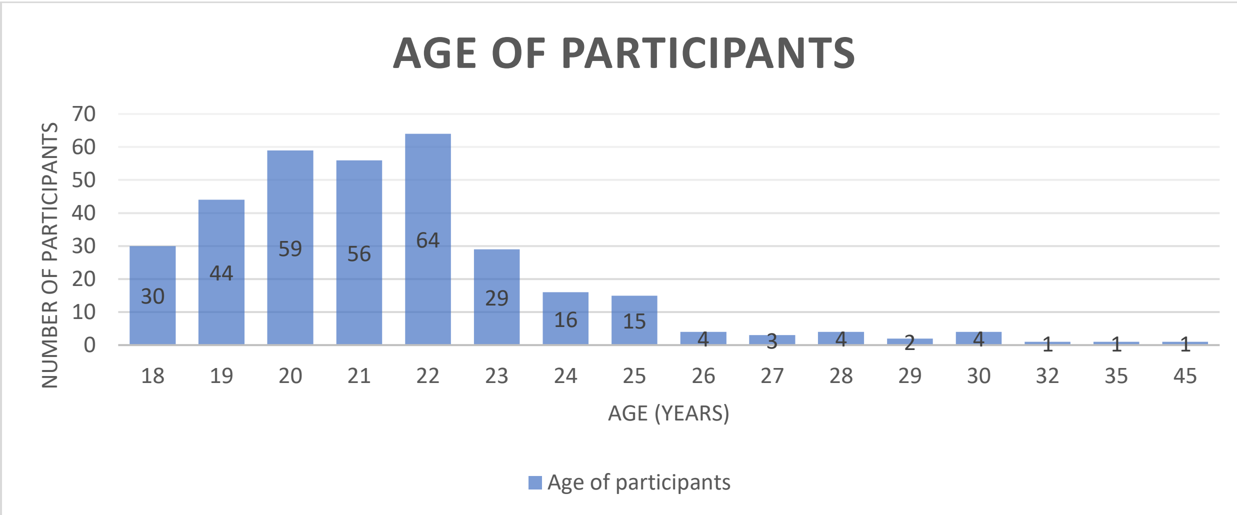


Table 5.2.1 – Gender

GENDER	Frequency	Percentage
Female	166	49.85
Male	163	48.95
Prefer not to say	4	1.2
Total	333	100

Table 5.2.2 – Level of study

LEVEL OF STUDY	Frequency	Percentage
100	92	27.63
200	70	21.02
300	59	17.72
400	87	26.13
500	4	1.2
600	4	1.2
POSTGRADUATE	17	5.11
Total	333	100

Table 5.2.3 – Colleges

COLLEGE	Frequency	Percentage
AGRICULTURE AND NATURAL RESOURCES	19	5.71
ARTS AND BUILT ENVIRONMENT	25	7.51
ENGINEERING	33	9.91
HEALTH SCIENCES	134	40.24
HUMANITIES AND SOCIAL SCIENCES	64	19.22
SCIENCE	58	17.42
Total	333	100

Figure 5.2.2 – Relationship Status

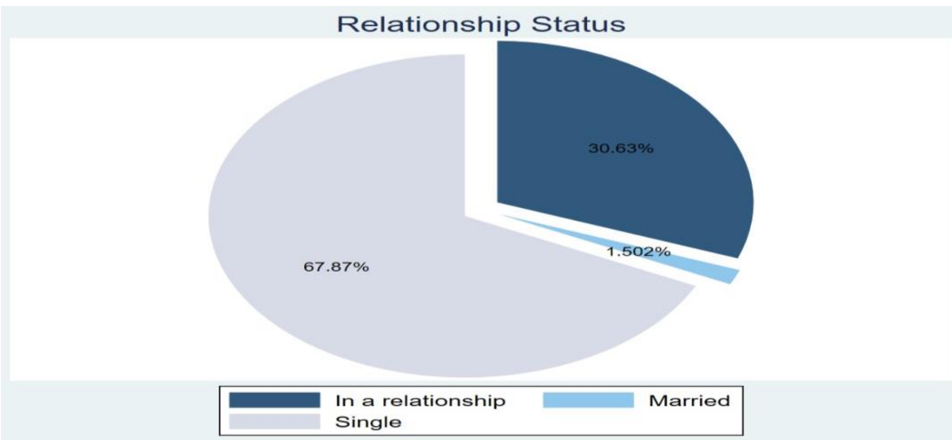


Figure 5.2.3 – Current Sexual Activity Status

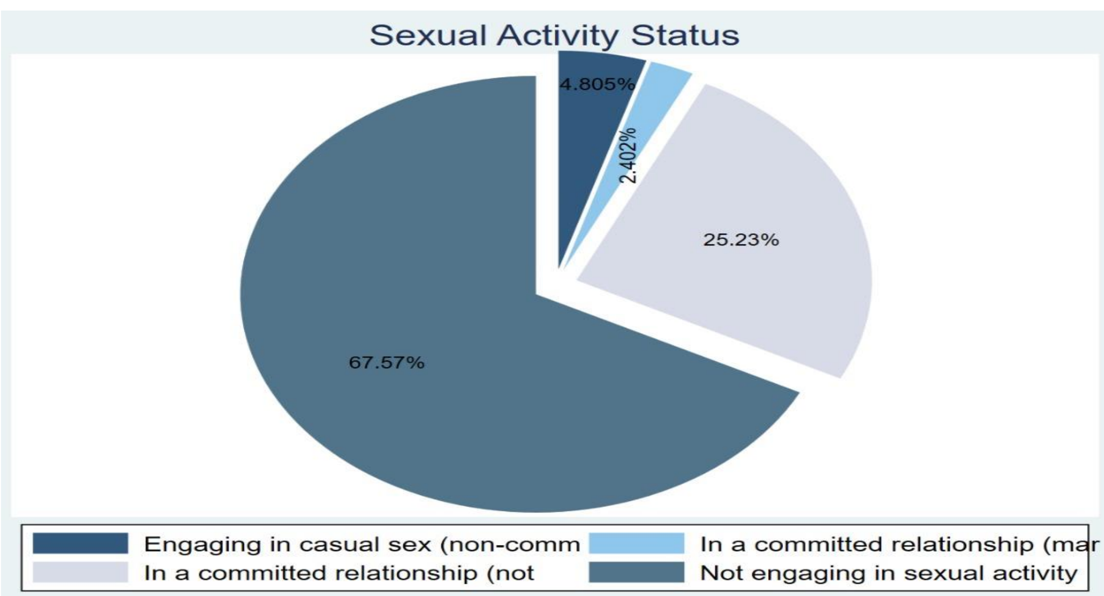


Figure 5.2.4 – Sexual Partners in last 2 years

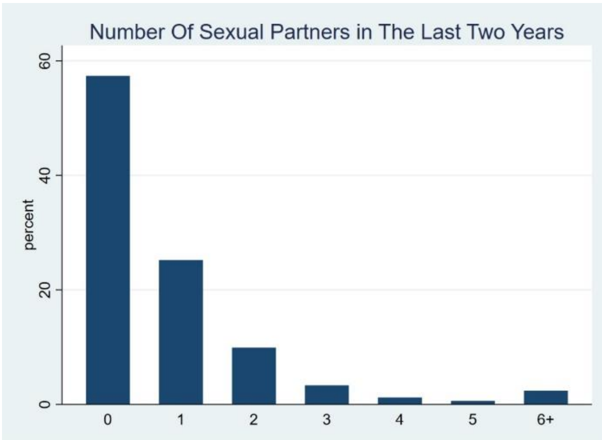
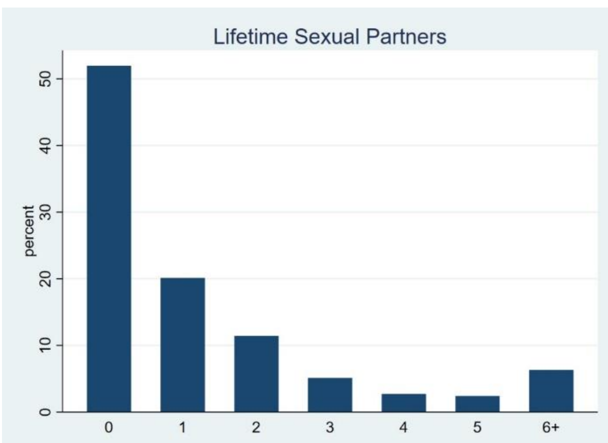


Figure 5.2.5 – Lifetime sexual partners



5.3 SOURCE OF INFORMATION ON SEXUALLY TRANSMITTED INFECTIONS

The survey results reveal a contrast between the sources people trust and those they actually use for STI information. While healthcare professionals (78.5%) and school/university courses (49.1%) are the most trusted, social media (88%) and internet sources (68.4%) are the most commonly used. Television/radio health talks and healthcare professionals are both widely used and trusted, suggesting their strong influence. In contrast, personal networks like friends, family, and traditional media are less trusted and used.

Figure 5.3.1 – Where do you primarily get information about STIs

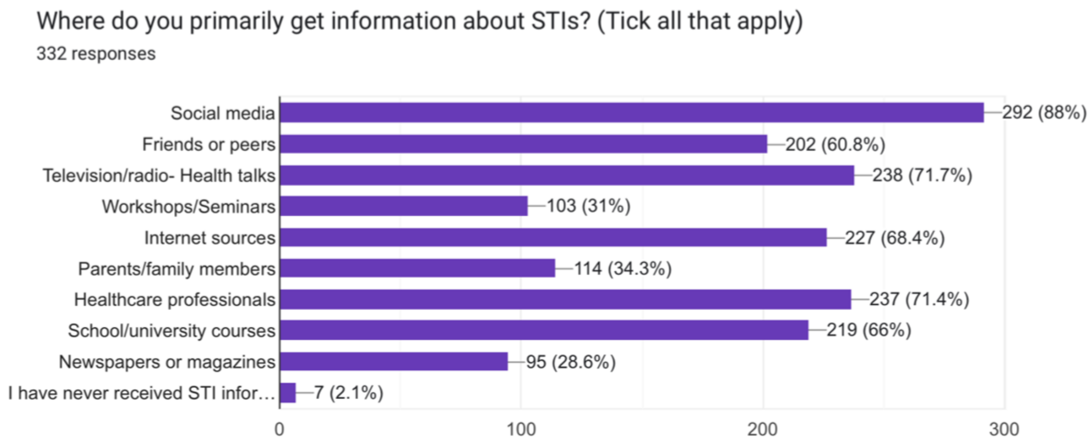
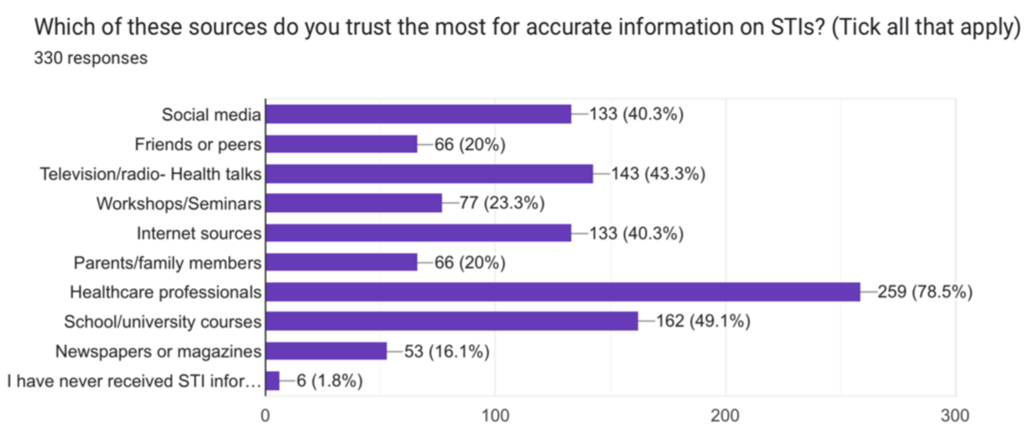


Figure 5.3.2 – Which of these sources do you trust most for accurate information on STIs?



5.4 AWARENESS OF SEXUALLY TRANSMITTED INFECTIONS

The survey data reveals varying levels of public awareness regarding different sexually transmitted infections (STIs). HIV/AIDS is the most recognized STI, with 99.4% of respondents indicating familiarity. Syphilis (93.9%) and Gonorrhea (93.3%) also show high levels of awareness, followed by Hepatitis B (88.4%). Chlamydia and Herpes were recognized by 73.9% and 70.5% of participants, respectively. In contrast, awareness drops significantly for Human Papilloma Virus (HPV) at 57.4%, and even more so for Trichomoniasis, which is known by only 47.4% of respondents. These results suggest strong general awareness of more widely discussed STIs, while highlighting a need for increased education around lesser-known but common infections like HPV and Trichomoniasis.

Figure 5.4.1 – *Awareness of various STIs*

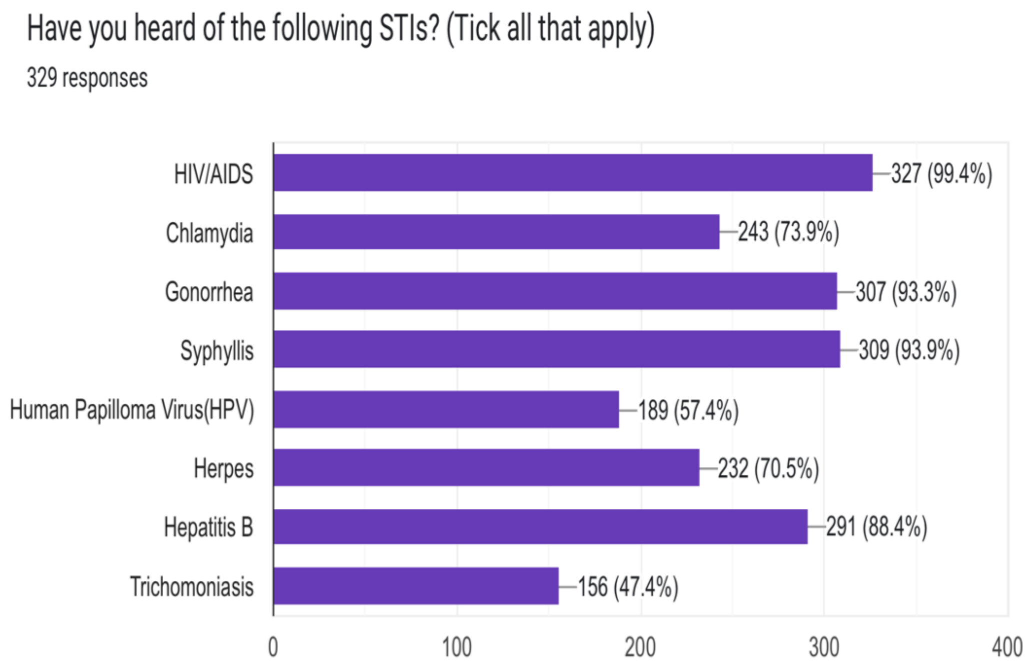
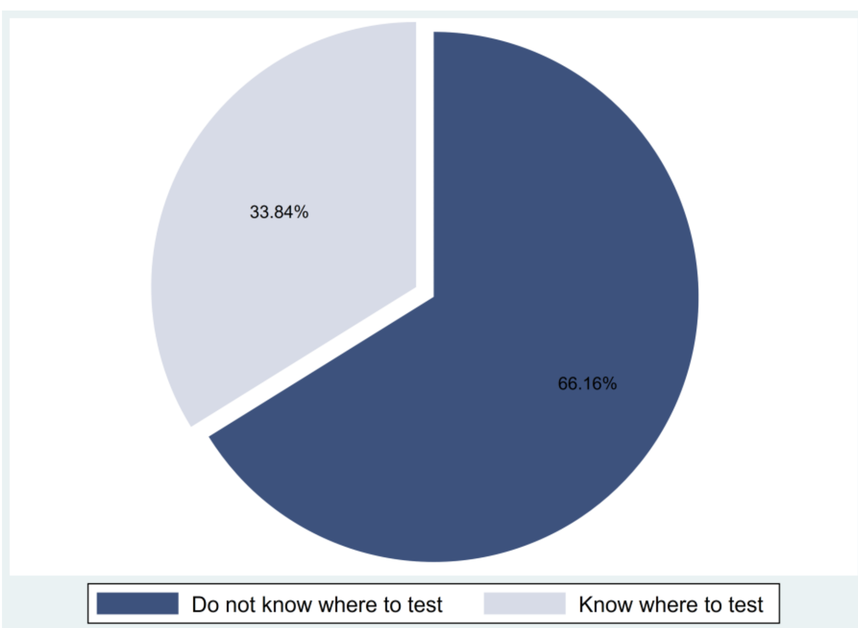


Figure 5.4.2 - *Awareness of STI testing services among KNUST students*.



5.5 KNOWLEDGE ON SEXUALLY TRANSMITTED INFECTIONS
5.5.1. TRANSMISSION
5.5.1.1 MODE OF TRANSMISSION OF SYPHILLIS

Most respondents (94.46%) correctly identify unprotected sexual contact as the primary mode of syphilis transmission, showing strong awareness. However, knowledge decreases for less obvious routes: 70.46% recognize kissing someone with a syphilitic sore as a risk, while only 51.99% are aware of mother-to-fetus transmission during pregnancy. Many mistakenly believe sharing toothbrushes or razors (68.73%) and sitting on public toilets (47.08%) can spread syphilis, reflecting common misconceptions. Overall, while baseline understanding of sexual transmission is high, significant gaps and myths remain, especially regarding vertical transmission and indirect contact, highlighting the need for targeted education to improve accurate knowledge and prevention.

Table 5.5.1 – Mode of transmission of syphilis

Mode of Transmission	% Correct Awareness	% Incorrect Awareness	% Uncertainty	Key Insight
Unprotected sexual contact (vaginal, oral, anal)	94.46% (Yes)	1.23% (No)	4.31% (Maybe)	Strong baseline knowledge of primary transmission route.
Kissing someone with a syphilitic sore	70.46% (Yes)	16.62% (No)	12.92% (Maybe)	Moderate awareness; some doubt about transmission through sores.
Sharing toothbrushes or razors	31.27% (No)	68.73% (Yes)	13.00% (Maybe)	Overestimation of risk; rare transmission but common misconception.
Mother to fetus during pregnancy	51.99% (Yes)	22.32% (No)	25.69% (Maybe)	Significant knowledge gap on vertical transmission.
Sitting on public toilets (not a mode of transmission)	32.92% (No)	47.08% (Yes)	20.00% (Maybe)	Misunderstanding about non-sexual transmission; myth prevalent.

5.5.1.2 MODE OF TRANSMISSION OF CHLAMYDIA

The data reveals that while a majority (87.7%) correctly identify unprotected sex as the main mode of Chlamydia transmission, significant misconceptions persist. Less than half (42.7%) are aware it can be passed from mother to baby during childbirth, and many incorrectly believe it can be transmitted through non-sexual means such as sitting on public toilets (47.6%), touching infected clothing (11.7%), or sharing food and drinks (10.3%). These misunderstandings suggest that public perception often overemphasizes casual contact risks while underestimating real sexual and perinatal transmission routes. This can lead to misguided behaviors, stigma, and missed opportunities for prevention, particularly during pregnancy. Targeted education is therefore essential to correct these beliefs and promote informed sexual health practices.

Table 5.5.2 – Mode of transmission of chlamydia

Mode of Transmission	% Correct Awareness	% Incorrect Awareness	% Uncertainty	Key Insight
Unprotected vaginal, oral, or anal sex	87.73% (Yes)	3.07% (No)	9.2% (Maybe)	High public awareness of the primary mode of Chlamydia transmission.
Sitting on public toilets	27.3% (No)	47.55% (Yes)	25.15% (Maybe)	Widespread misconception; many wrongly believe Chlamydia spreads via toilets.
Mother-to-child during childbirth	42.68% (Yes)	27.1% (No)	30.22% (Maybe)	Low awareness of vertical transmission; risk of missed prenatal care.
Touching infected clothing	64.2% (No)	11.73% (Yes)	24.07% (Maybe)	Moderate understanding; over one-third

				still misinformed or unsure.
Sharing food or drinks	69.57% (No)	10.25% (Yes)	20.19% (Maybe)	Most recognize this is not a transmission route, though some confusion remains.

5.5.1.3 MODE OF TRANSMISSION OF HIV/AIDS

The analysis reveals that while most respondents accurately recognize high-risk transmission routes such as unprotected sex (94.86%) and sharing needles (90.85%), misconceptions persist regarding non-transmissible or low-risk interactions. A strong majority correctly identified that HIV cannot be spread through hugging (94.19%) or mosquito bites (83.28%), though the latter still sees nearly 17% of respondents either misinformed or unsure. The most concerning finding is that only 26.06% correctly knew that kissing does not transmit HIV, while over 73% held incorrect beliefs or uncertainty. These results highlight a critical need for targeted public health education to dispel myths, particularly around casual contact, and to reduce stigma toward people living with HIV/AIDS.

Table 5.5.3 – Mode of transmission of HIV/AIDS

Mode of Transmission	% Correct Awareness	% Incorrect Awareness	% Uncertainty	Key Insight
Unprotected sex	94.86% (Yes)	3.02% (No)	2.11% (Maybe)	High public awareness of the most common mode of HIV transmission.
Mosquito bites	83.28% (No)	9.12% (Yes)	7.60% (Maybe)	Persistent myth; some still wrongly believe mosquitoes transmit HIV.
Sharing of needles	90.85% (Yes)	4.57% (No)	4.57% (Maybe)	Strong understanding of this high-risk transmission route.

Hugging someone with HIV/AIDS	94.19% (No)	2.14% (Yes)	3.67% (Maybe)	Most understand HIV isn't spread through casual physical contact.
Kissing someone with HIV/AIDS	26.06% (No)	50.61% (Yes)	23.33% (Maybe)	Major misconception; majority falsely believe kissing can transmit HIV.

5.5.1.4 MODE OF TRANSMISSION OF GONORRHEA

The findings show that while the majority of respondents (97.28%) correctly identified unprotected sexual activity as a mode of gonorrhea transmission, significant misconceptions persist regarding other routes. Over half (57.8%) mistakenly believe that gonorrhea can be spread through sharing toilet seats, and 17.13% think it can be transmitted by wearing infected clothes. Additionally, awareness of mother-to-child transmission during childbirth is limited, with only 52.44% answering correctly. These results highlight a need for targeted education to address myths about casual contact and improve understanding of less commonly known transmission routes like vertical transmission.

Table 5.5.4 – Mode of transmission of Gonorrhea

Mode of Transmission	% Correct Awareness	% Incorrect Awareness	% Uncertainty	Key Insight
Unprotected sexual contact (vaginal, oral, anal)	97.28% (Yes)	1.21% (No)	1.51% (Maybe)	Strong understanding of primary transmission route.
Using the same toilet seat	30.58% (No)	57.8% (Yes)	11.62% (Maybe)	Common myth; majority incorrectly believe in transmission via toilet seats.
From mother to baby during childbirth	52.44% (Yes)	28.05% (No)	19.51% (Maybe)	Moderate awareness; notable gap in understanding

				vertical transmission.
Wearing infected clothes	68.5% (No)	17.13% (Yes)	14.37% (Maybe)	Fair awareness; some misconceptions about transmission through clothing.

5.5.2. SYMPTOMS

The survey data reveals varied levels of public awareness regarding common symptoms of gonorrhea, chlamydia, syphilis, and HIV. Awareness is highest for gonorrhea and HIV, with 86.77% and 76.99% of respondents correctly identifying pain during urination and weight loss as key symptoms, respectively. In contrast, knowledge gaps are more pronounced for chlamydia and syphilis, where nearly a quarter or more of respondents admitted they did not know the common symptoms. Small but consistent misinformation is present across all infections, with some respondents selecting unrelated symptoms.

These findings highlight the need for targeted sexual health education, particularly around chlamydia and syphilis, to address the significant uncertainty and correct misconceptions. While gonorrhea and HIV symptoms are relatively well recognized, ongoing public health efforts should reinforce this knowledge to ensure timely diagnosis and treatment. Overall, tailored awareness campaigns focusing on lesser-known symptoms and clarifying myths could help reduce the transmission and health impacts of these infections.

Table 5.5.5 – Symptoms of Sexually Transmitted Infections

Infection	Hallmark (h.) Symptom	Mild Symptoms (Other Recognized)	Wrong Symptoms (Incorrect Answers)	% Correct h. symptom	% Incorrect h. symptom	% Clueless	Key Insight
Gonorrhea	Pain during urination (86.77%)	–	Coughing, rash on hands (3.38%)	86.77%	3.38%	9.85%	High awareness; small misinformation; most respondents correctly

							identified main symptom.
Chlamydia	Vaginal/penile discharge (72.84%)	–	Fever, headache, vomiting (3.39%)	72.84%	3.39%	23.77%	Moderate awareness with high uncertainty; targeted education needed to improve knowledge.
Syphilis	Painless sore (chancre) (68.52%)	–	Itchy eyes, stomach pain (4.32%)	68.52%	4.32%	27.16%	Lowest correct recognition; highest uncertainty; significant education gap on symptom knowledge.
HIV	Weight loss (76.99%)	Fatigue (13.5%), Night sweats (4.6%)	–	76.99%	~8.1%	4.91%	Strong recognition of hallmark symptom; fatigue and night sweats seen as mild; low uncertainty.

5.6 RISK PERCEPTION ON SEXUALLY TRANSMITTED INFECTIONS

The data shows that there is a widespread awareness of the behaviors and conditions that increase the risk of contracting sexually transmitted infections (STIs). An overwhelming 93% of respondents understand that having multiple sexual partners raises STI risk, and nearly 74% believe condom use is necessary even with trusted partners, reflecting a strong grasp of preventive measures. Additionally, more than half of the respondents perceive that students at KNUST are likely or very likely to be at risk of contracting STIs, indicating a general acknowledgment of the vulnerability within the community.

Despite this broad understanding of risk factors, personal risk perception appears considerably lower. Only about 19% of individuals consider themselves at risk of contracting an STI, while the majority (70%) believe they are not at risk. This disparity between general awareness and personal vulnerability highlights a common psychological gap where individuals acknowledge risks in theory but do not translate that awareness into their own context. Moreover, nearly 11% remain uncertain about their personal risk, which suggests an opportunity for targeted educational efforts to help people better assess their own behaviors and risks.

Concern about STIs is generally high among respondents, with about 65% expressing moderate to very high levels of concern. This concern aligns with the recognition of STI risks but contrasts with the lower personal risk perception, implying that while STIs are seen as serious public health issues, many individuals may not feel personally threatened. This pattern underscores the importance of risk communication strategies that move beyond knowledge transmission to address attitudes and behaviors that influence self-assessment of vulnerability.

Overall, the combined data highlights that while knowledge of STI risks and preventive behaviors is strong, the critical challenge lies in bridging the gap between general risk awareness and personal risk perception. Public health interventions should focus on personalized education, risk self-assessment, and promoting consistent protective practices even within trusted relationships to enhance effective STI prevention and control.

Figure 5.6.1 – Self-perceived risk of contracting an STI

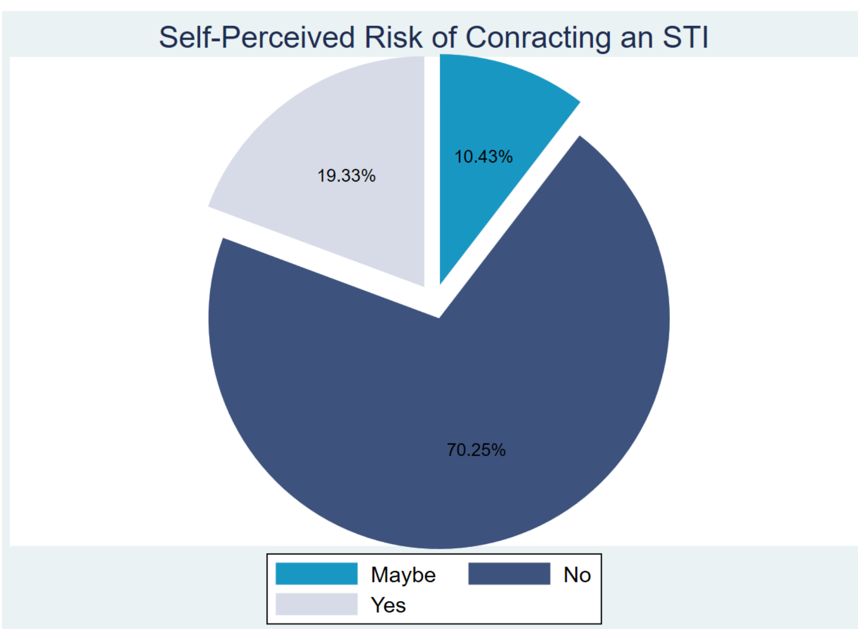


Table 5.6.1 – Risk Perception of Sexually Transmitted Infections

Aspect of Risk Perception	Response Highlights	Interpretation / Insight
Perceived personal risk of contracting an STI	Yes: 19.33%- No: 70.25%- Maybe: 10.43%	Majority (70%) do not perceive themselves at risk despite general STI awareness. Indicates low personal risk perception.
Concern level about STIs	- High concern (4 & 5): 65.03%- Low concern (1 & 2): 18.71%	While many express high concerns about STIs, this does not fully align with their personal risk perception.
Community-level risk perception (students at KNUST)	High concern (4 to 5): 57.97%, Low concern (1 to 2): 15.64	Majority perceive a high risk for the student population, showing awareness of broader environmental or peer risk.
Belief about multiple sexual partners increasing STI risk	- Yes: 93.27%- No: 4.28%- Maybe: 2.45%	Strong consensus on multiple partners as a major risk factor, reflecting good knowledge of behavioural risk factors.
Belief about condom use necessity with trusted partner	- Yes: 73.85%- No: 14.15%- Maybe: 12%	Majority advocate condom use even with trusted partners, showing recognition of ongoing risk despite trust.

5.7 PREVENTIVE BEHAVIORS ON SEXUALLY TRANSMITTED INFECTIONS

Among sexually active respondents, only 27.9% reported always using condoms, 47.9% used them sometimes, and 24.3% never used them. This highlights inconsistent condom use among students. Out of all respondents, 31.3% reported having ever been tested for a sexually transmitted infection (STI), while 68.7% had never undergone testing. Among those who had been tested, 21.6% reported receiving a positive STI diagnosis, compared to only 1.3% among those who had never been tested. Overall, 7.7% of participants reported testing positive for an STI. These findings suggest a strong association between STI testing and the likelihood of detecting an infection. The significantly lower positivity rate among those who have never been tested highlights the potential for underdiagnosis and emphasizes the importance of regular testing to ensure early detection and treatment.

Awareness of STI testing services is also limited, with two-thirds unaware of available resources on or near campus, likely contributing to the low testing rates. On a positive note, the majority (77.85%) expressed willingness to inform sexual partners if diagnosed with an STI, indicating a strong ethical attitude toward disclosure and prevention. However, about 22% remain uncertain or unwilling to disclose, suggesting the need for targeted education around communication and responsibility in sexual health. Overall, these findings underscore the urgent need for improved sexual health education, increased testing accessibility, and efforts to normalize conversations about STI prevention and disclosure.

Figure 5.7.1 – Condom use consistency among sexually active students

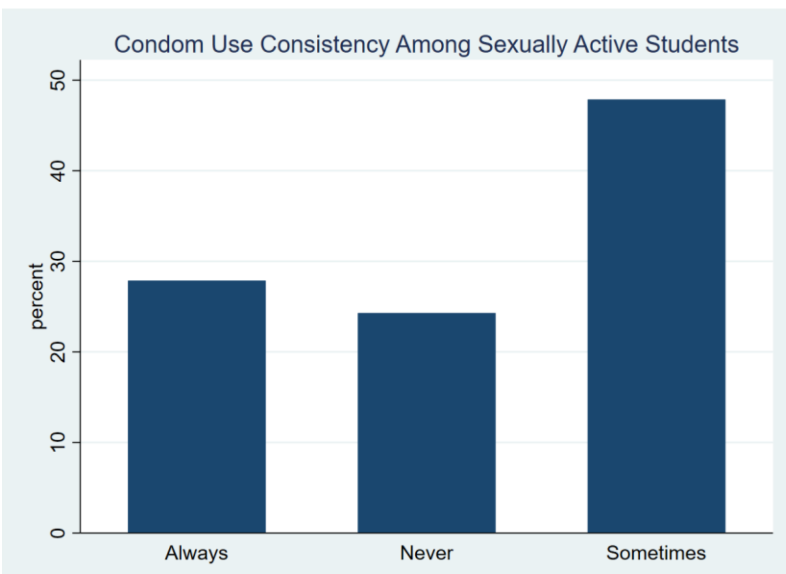


Figure 5.7.2 – Bar chart of predictors of STI testing behaviour.

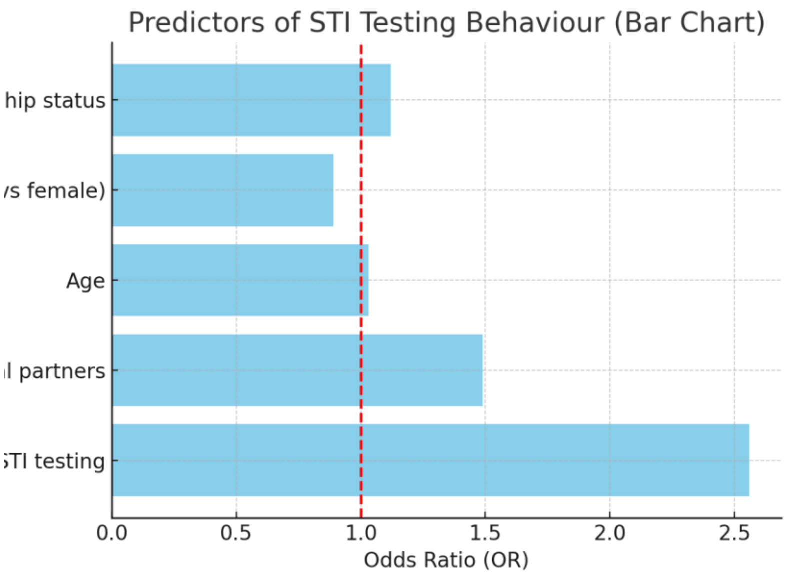


Figure 5.7.3 – Forest plot of predictors of STI testing behaviour.

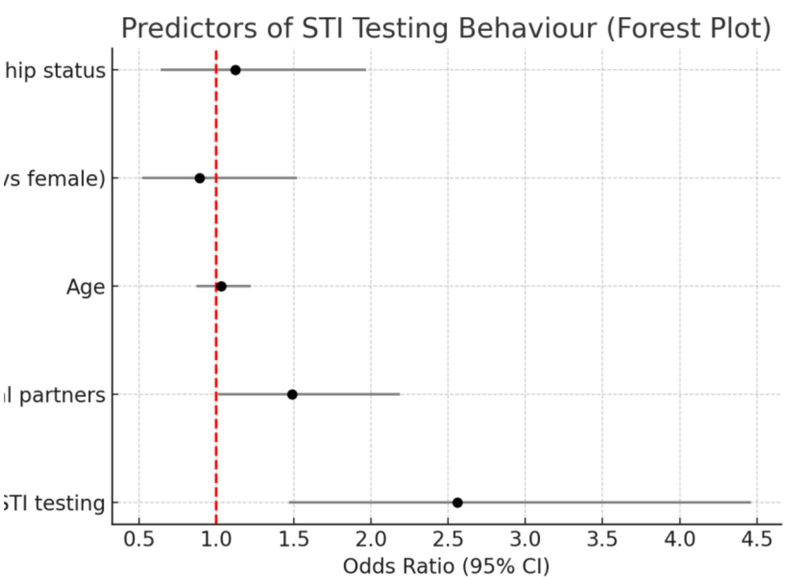


Table 5.7.1 – Preventive behaviors of Sexually Transmitted Infections

Behaviour / Indicator	Response Category	Percent (%)	Key Insights
Consistent Condom Use	Always	12.00%	Very low consistent use; most sexually active respondents engage in risk.
	Sometimes / Never	31.08%	Inconsistent or no use puts individuals at higher risk of STIs or pregnancy.
	Not Sexually Active	56.92%	Majority are not currently sexually active, lowering immediate risk.
Ever Tested for STIs	Yes	31.19%	Less than one-third have ever been tested, indicating poor preventive engagement.
	No	68.81%	Majority have never been tested, raising concern about undiagnosed STIs.
Tested Positive for STI	Yes	7.65%	Indicates STI presence; among tested individuals, 1 in 4 were positive.
Awareness of STI Testing Services on/near Campus	Yes	33.84%	Low awareness may limit access and contribute to low testing rates.
	No	66.16%	High information gap; many unaware of available health services.
Willingness to Inform Partner if Diagnosed with STI	Yes	77.85%	Strong sense of ethical responsibility among most respondents.
	Maybe / No	22.15%	Some uncertainty or refusal to disclose, risking further transmission.

5.8 DETERMINANTS OF STUDENTS' AWARENESS, RISK PERCEPTION AND PREVENTIVE BEHAVIORS

Logistic regression results showed that age ≥21 years, awareness of STIs, consistent condom use, and perceiving oneself at risk were significant predictors of STI testing behaviour. Gender, however, was not statistically significant. This indicates that behavioural and cognitive factors play a stronger role than sex differences in influencing STI testing uptake.

Table 5.8.1. Determinants of STI Testing Behaviour

Variable	**Odds Ratio (OR)**	**95% CI (Lower–Upper)**	**p-value**
Age (≥ 21 years)	1.8	1.1 – 2.9	0.021
Gender (Male)	0.7	0.4 – 1.2	0.182
Awareness of STIs	2.5	1.3 – 4.8	0.006
Condom use (Always)	1.9	1.0 – 3.4	0.047
Perceived risk (Yes)	2.2	1.2 – 4.1	0.011

Logistic regression results for determinants of STI testing behaviour among students.

Cross-tabulation of Risk Perception and Gender

A greater proportion of males (21.74%) reported perceiving themselves at risk of contracting STIs compared to females (17.39%). This suggests that male students may be more aware of their vulnerability, potentially due to higher engagement in risk behaviours.

Table 5.8.2. Cross-tabulation of Risk Perception and Gender

Gender	**At Risk(%)**	**Not at risk(%)**	**Maybe(%)**
Male	21.74	63.98	14.29
Female	17.39	75.78	6.83

Distribution of students' self-perceived STI risk by gender.

Cross-tabulation of Risk Perception and Age Group

Older students (≥25 years) were more likely to perceive themselves at risk (28.57%) compared to those aged ≤20 years (18.11%). This indicates that risk perception increases with age, possibly reflecting greater sexual experience or exposure to health education.

Table 5.8.3. Cross-tabulation of Risk Perception and Age Group

Age Group	**At Risk(%)**	**Not at Risk(%)**	**Maybe (%)**
≤ 20 yrs	18.11	75.59	6.3
21–24 yrs	18.29	67.68	14.03
≥ 25 yrs	28.57	62.86	8.57

Distribution of students' self-perceived STI risk by age group.

Cross-tabulation of Risk Perception and Sexual Activity

Sexually active students reported much higher levels of perceived risk (35.2%) compared to those who were not sexually active (8.1%). This suggests that engaging in sexual activity increases awareness of vulnerability to STIs, though a sizable portion of sexually active students still do not perceive risk.

Table 5.8.4. Cross-tabulation of Risk Perception and Sexual Activity

Sexual Activity	**At Risk(%)**	**Not at Risk(%)**	**Maybe (%)**
Engaging in casual sex (not-committed)	56.25	37.50	6.25
In a committed relationship (married)	37.50	50.00	12.5
In a committed relationship (not married)	22.89	62.65	14.46
Not engaging in sexual activity	14.61	76.26	9.13

Distribution of students' self-perceived STI risk by sexual activity status.

The findings reveal that while general awareness of STI transmission through unprotected sex is impressively high among KNUST students; 94.46% for syphilis, 94.86% for HIV/AIDS, and 97.28% for gonorrhea, there are notable gaps in knowledge related to less obvious modes of transmission and symptom recognition. This reflects the influence of **knowledge depth** as a determinant. Only 51.99% recognized vertical transmission of syphilis, 52.44% for gonorrhea, and 42.7% for chlamydia. Additionally, while awareness of common symptoms was strong for gonorrhea (86.77% identified pain during urination) and HIV/AIDS (76.99% identified weight loss), fewer students were able to identify symptoms of syphilis (59.01%) and chlamydia (66.76%). These findings suggest that students possess strong surface-level understanding of STIs but lack the **comprehensive knowledge** required to fully assess personal risk or identify infections that may present with mild or no symptoms.

Furthermore, the study revealed widespread **misconceptions and inaccurate beliefs** about how STIs are transmitted. A significant number of students believed some STIs could be contracted through non-sexual means; 68.73% believed syphilis could be spread through sharing razors or toothbrushes, 57.8% thought gonorrhea could be contracted from toilet seats, and nearly half believed that sitting on public toilets could transmit chlamydia (47.6%) or syphilis (47.08%). While **healthcare professionals (78.5%)** and **school or university courses (49.1%)** were reported as the most **trusted sources** of STI information, the most frequently used sources were **social media (88%)** and **internet sources (68.4%)**. This contrast highlights the importance of **access to credible information** as a key determinant of STI awareness and behavior. The disconnect between trusted and commonly used sources emphasizes the urgent need to ensure that digital platforms, especially social media, deliver accurate and accessible sexual health information, as they remain the primary channels through which students engage with STI-related content.

Individuals engaging in casual, non-committed sex reported the highest perception of risk, with 56.25% identifying themselves as "At Risk." In contrast, those not engaging in sexual activity overwhelmingly perceived themselves as "Not at Risk" (76.26%), with only 14.61% considering themselves "At Risk." Among individuals in committed relationships, perceptions varied: Those married showed a moderate level of risk awareness, with 37.5% feeling "At Risk" and 50% "Not at Risk." Those in non-marital committed relationships were more likely to perceive low risk, with 62.65% identifying as "Not at Risk" and only 22.89% as "At Risk." The "Maybe" category was most prominent among individuals in committed relationships, suggesting some uncertainty about STI risk even within seemingly stable partnerships.

CHAPTER SIX

DISCUSSION

6.1 INTRODUCTION

This chapter seeks to elaborate on the findings of the study. The research on awareness, risk perception, and preventive measures for sexually transmitted infections (STIs) among students at Kwame Nkrumah University of Science and Technology sheds critical light on the existing knowledge gaps, attitudes, and behaviors influencing STI prevention in a university setting. Understanding students' level of awareness and how they perceive their risk is essential for tailoring effective educational and health interventions. The findings highlight not only the extent to which students are informed about STIs but also how their perceptions impact their engagement with preventive practices such as condom use and regular testing. By exploring these dimensions, the study contributes valuable insights into the challenges and opportunities for enhancing sexual health programs aimed at reducing STI transmission within this youthful and academically diverse population.

6.2 AWARENESS OF SEXUALLY TRANSMITTED INFECTIONS

This study found HIV/AIDS to be the most widely recognized STI, with 99.4% of respondents indicating awareness. This aligns with earlier studies showing that HIV/AIDS consistently ranks highest in public knowledge due to sustained global and national awareness campaigns (Adih & Alexander, 1999; Agyemang et al., 2012). Syphilis (93.9%) and gonorrhea (93.3%) were also highly recognized, reflecting their long-standing presence and visibility within both medical discourse and community health education (Anarfi, 2005). Awareness of Hepatitis B was relatively high at 88.4%, a finding consistent with prior reports that identified growing recognition of viral hepatitis as a public health issue in Ghana (Abdulai et al., 2008). In contrast, knowledge of Chlamydia (73.9%) and Herpes (70.5%) was lower, echoing earlier research that such infections tend to receive less emphasis in health promotion efforts compared to HIV and gonorrhea (Anarfi, 2005). Notably, Human Papilloma Virus (HPV) and trichomoniasis recorded the lowest levels of awareness, at 57.4% and 47.4% respectively. Similar gaps have been observed in previous studies, where poor recognition of HPV was linked to limited health education and the relatively recent introduction of HPV vaccination programs (Abotchie & Shokar, 2009). Trichomoniasis, despite its prevalence, remains under-recognized largely due to minimal health communication surrounding the condition (Mayaud & Mabey, 2004). Taken together, these findings demonstrate the effectiveness of awareness campaigns in promoting knowledge of the more common STIs, while emphasizing the need to expand educational efforts to include less-discussed but equally significant infections such as HPV and trichomoniasis.

6.3 KNOWLEDGE ON SEXUALLY TRANSMITTED INFECTIONS

The findings indicate that students at Kwame Nkrumah University of Science and Technology (KNUST) demonstrate a high level of awareness regarding the sexual transmission of common sexually transmitted infections (STIs). Over 94% of respondents correctly identified unprotected sex as a mode of transmission for syphilis (94.46%), HIV/AIDS (94.86%), and gonorrhea (97.28%), while 87.7% recognized it as a transmission route for chlamydia. This strong awareness mirrors data from other university settings; for instance, a study conducted at the University of Gondar in Ethiopia reported that 90.4% of students identified unprotected sex as the primary route of STI transmission (Wegene et al., 2019). Such widespread awareness suggests that core messaging around sexual transmission has been relatively effective in university contexts.

However, despite the high awareness of sexual transmission routes, significant knowledge gaps remain, particularly regarding vertical (mother-to-child) transmission. In this study, only about half of respondents recognized vertical transmission as a possible route for syphilis (51.99%), gonorrhea (52.44%), and chlamydia (42.7%). This gap in knowledge is concerning, as vertical transmission can lead to severe neonatal complications if undiagnosed and untreated. Similar findings have been reported in other settings, such as in Togo and Ethiopia, where university students demonstrated poor understanding of non-sexual STI transmission routes, including vertical transmission (Chemerinski et al., 2020; Tchao et al., 2021). These results emphasize the need to extend STI education beyond sexual contact alone.

Another major area of concern is the persistence of misconceptions about STI transmission through casual contact. A significant proportion of students believed STIs could be contracted through non-sexual means: 68.73% believed syphilis could be spread via sharing razors or toothbrushes, 57.8% believed gonorrhea could be transmitted through toilet seats, and nearly half believed chlamydia (47.6%) and syphilis (47.08%) could be acquired from sitting on public toilets. These misconceptions are not isolated to KNUST students. Misinformation about STI transmission through toilet seats or shared surfaces has been widely reported in youth-focused media and public discourse (Schreiber, 2022; Teen Vogue, 2017). Unfortunately, such beliefs can perpetuate unnecessary stigma, as individuals wrongly associate STI contraction with poor hygiene rather than specific risk behaviors (Jabeen et al., 2022; Legro et al., 2013).

Symptom recognition is another dimension where knowledge levels vary. Awareness is highest for gonorrhea, where 86.77% of students correctly associated the infection with pain during urination, and HIV/AIDS, with 76.99% recognizing weight loss as a symptom. However, recognition of symptoms for syphilis and chlamydia is considerably weaker, with over 25% of students admitting they did not know the symptoms. These results are consistent with prior research from South Africa, where many university students acknowledged that STIs can be asymptomatic, yet lacked detailed knowledge of specific symptoms (Mhlongo et al., 2021). Similarly, Lorimer and Hart (2010) found that while awareness of chlamydia as an STI was relatively high, understanding of its asymptomatic nature and potential complications was limited. Such gaps in symptom knowledge may contribute to delayed testing and treatment, especially for infections that progress silently.

Notably, gender differences emerged in both knowledge and health-seeking behavior. Female students tended to demonstrate higher levels of STI awareness and were more likely to engage in screening and preventive health services. This aligns with findings from the University of Kara in Togo, where STI knowledge was significantly associated with being female, having higher education levels, and having previously undergone HIV testing (Tchao et al., 2021). Other studies suggest that while males may exhibit stronger knowledge around condom use, females are more proactive in seeking health information and utilizing care services (Greaves et al., 2009; Pleck et al., 1992). These disparities underscore the need for gender-sensitive approaches to STI education and prevention, ensuring that interventions reach all subgroups effectively.

Overall, while knowledge of sexual transmission routes is relatively strong among KNUST students, critical gaps persist in other areas, including awareness of vertical transmission, symptom recognition, and the debunking of common myths. Addressing these gaps requires comprehensive, culturally adapted, and peer-led sexual health education programs. Interventions that incorporate digital platforms, myth-busting campaigns, and gender-specific strategies are likely to have the greatest impact in bridging these knowledge deficits and promoting informed sexual health behaviors.

6.4 RISK PERCEPTION ON SEXUALLY TRANSMITTED INFECTIONS

The results show that while awareness of risk factors for sexually transmitted infections (STIs) is high, personal risk perception remains relatively low. A large majority of respondents (93.3%)

correctly identified having multiple sexual partners as a major risk factor, while 73.9% acknowledged the importance of condom use even within trusted relationships. This aligns with previous studies which found that knowledge of STI transmission and preventive strategies is generally high among university students (Oppong Asante, 2013; Amu & Adomah-Afari, 2016). Furthermore, more than half of respondents (57.9%) believed students at KNUST are at high risk of contracting STIs, suggesting recognition of community-level vulnerability.

Despite this, only 19.3% considered themselves personally at risk, whereas 70.3% reported no risk, and 10.4% remained uncertain. This discrepancy between general risk acknowledgment and personal susceptibility mirrors findings from earlier research, where students underestimated their individual vulnerability despite recognizing risk at a population level (Eaton et al., 2012; Olapegba, 2015). Such a psychological gap—often explained by the "optimistic bias"—has been reported widely, with individuals perceiving themselves as less likely to contract STIs than their peers (Weinstein, 1987; Mavhandu-Mudzusi & Asgedom, 2016).

Additionally, concern about STIs was moderate to high among 65.0% of respondents, indicating that while the infections are seen as serious public health issues, concern does not necessarily translate into heightened self-risk appraisal or consistent protective behavior. This echoes previous studies which noted that concern and knowledge alone are insufficient drivers of preventive practices (Njau et al., 2014; Crosby et al., 2018).

Overall, the findings emphasize that the main challenge is not awareness, but the translation of that awareness into accurate personal risk perception and preventive action. Public health interventions should therefore focus on personalized risk communication and self-assessment strategies that encourage consistent condom use, even within trusted relationships, as this remains a critical gap in STI prevention among young adults in university settings (Oppong Asante, 2013; Amu & Adomah-Afari, 2016).

6.5 PREVENTIVE BEHAVIORS OF SEXUALLY TRANSMITTED INFECTIONS

As shown in Figure 5.7.1, condom use among sexually active respondents was inconsistent, with only 27.9% reporting "always" using condoms, while nearly three-quarters used them irregularly or not at all. This exposes students to significant risk despite high awareness of STIs reported earlier (see Sections 5.4–5.6). Similar findings of low consistent use among African students have been reported (Amu & Adegun, 2015; Oppong Asante, 2013).

Testing uptake was also low: only 31.3% had ever tested for an STI, though among them, 21.6% reported a positive diagnosis (Table 5.7.1). This suggests underdiagnosis among the majority who had never tested, consistent with studies in Ghanaian universities (Eliason et al., 2014; Tabong et al., 2018).

Figures 5.7.2 and 5.7.3 highlight predictors of testing. Awareness of available services and higher perceived risk were the strongest motivators. Females were more likely to test than males, aligning with gendered differences in health-seeking behaviour (Mogaji et al., 2019). Interestingly, consistent condom users were less likely to test, possibly due to a perceived sense of protection.

Awareness of on-campus testing services was low (33.8%), which likely explains poor uptake. Positively, most respondents (77.9%) were willing to disclose an STI diagnosis to partners, though about one in five remained unwilling—an issue linked to stigma and fear of rejection (WHO, 2021).

Overall, these findings show a gap between knowledge/risk perception and preventive practice. Improving awareness of services, ensuring access, and normalizing routine testing are critical for reducing STI transmission among students.

6.6 DETERMINANTS OF STUDENTS' AWARENESS, RISK PERCEPTION AND PREVENTIVE BEHAVIORS

This study investigated the determinants of sexually transmitted infection (STI) testing behavior among students at KNUST, revealing that behavioral and cognitive factors—namely age, awareness of STIs, consistent condom use, and perceived risk—were significant predictors of testing uptake. These findings align with regional evidence that emphasizes the role of individual-level awareness and perceived vulnerability in shaping sexual health behaviors (Antwi et al., 2022; Atupra et al., 2021).

The logistic regression results showed that students aged 21 years and above were significantly more likely to undergo STI testing. This may reflect increased sexual experience and exposure to health education, as older students are more likely to have encountered STI-related messaging through formal or informal channels. Similar age-related trends were observed in studies conducted in Sunyani and Accra, where older adolescents demonstrated higher STI risk perception and testing intentions (Antwi et al., 2022; Abroso, 2013).

Awareness of STIs emerged as a strong predictor of testing behavior, consistent with findings from the University of Ghana and Cape Coast, where knowledge about STIs was high but not always matched by preventive practices (Acheampong, 2017; Atupra et al., 2021). This gap between knowledge and action underscores the need for interventions that go beyond information dissemination to address behavioral intentions and perceived susceptibility.

Condom use was also positively associated with STI testing, suggesting that students who engage in protective sexual practices may be more health-conscious and proactive in seeking testing. This is supported by regional data indicating that condom use among sexually active youth correlates with higher health literacy and STI awareness (James et al., 2022). However, the relatively modest odds ratio suggests that condom use alone may not be sufficient to drive testing behavior without accompanying risk perception.

Perceived risk was a significant determinant, reinforcing the Health Belief Model's assertion that individuals are more likely to engage in preventive health behaviors when they recognize their vulnerability (Sunyani Study, 2022). Cross-tabulation results showed that sexually active students, particularly those engaging in casual sex, reported higher levels of perceived risk. Yet, a substantial proportion of sexually active students still did not perceive themselves at risk, echoing findings from Ghanaian studies that highlight low risk perception despite high engagement in risky behaviors (Barimah et al., 2022).

Interestingly, gender was not a statistically significant predictor of STI testing, diverging from some earlier studies in Ghana that reported higher testing rates among females (Acheampong, 2017). This may reflect shifting norms and increased sexual health engagement among male students, as suggested by the higher proportion of males in this study who perceived themselves at risk.

Overall, the findings suggest that STI testing behavior at KNUST is more strongly influenced by cognitive and behavioral factors than by demographic characteristics. Health promotion strategies should therefore focus on enhancing STI awareness, fostering accurate risk perception, and promoting consistent condom use. Tailored interventions that address the specific needs and behaviors of sexually active youth—especially those in casual relationships—are essential for improving testing uptake.

CHAPTER SEVEN

CONCLUSION AND RECOMMENDATIONS

7.1 CONCLUSION

This study explored the determinants of STI testing behavior among students at Kwame Nkrumah University of Science and Technology, highlighting the critical role of behavioral and cognitive factors over demographic characteristics. Logistic regression analysis revealed that age $\geqslant 21$ years, awareness of STIs, consistent condom use, and perceived risk were significant predictors of testing uptake. Gender, however, was not statistically significant, suggesting that sex-based differences may be less influential in this context.

Cross-tabulation findings further emphasized the importance of perceived risk, which was notably higher among older students and those engaging in sexual activity—particularly casual sex. Despite this, a substantial proportion of sexually active students did not perceive themselves at risk, pointing to a disconnect between behavior and risk awareness. This gap presents a key opportunity for targeted interventions.

The findings underscore the need for comprehensive sexual health education that goes beyond basic awareness to foster accurate risk perception and promote proactive health-seeking behavior. Programs should be tailored to address the unique needs of sexually active youth, especially those in non-committed relationships, and should aim to normalize STI testing as a routine aspect of sexual health.

Future research should consider longitudinal designs to assess changes in behavior over time and explore additional factors such as stigma, accessibility of testing services, and peer influence. Strengthening STI prevention efforts within tertiary institutions will be essential to safeguarding the health and well-being of Ghana's youth.

7.2 RECOMMENDATIONS

Based on the results of this study, there is a pressing need for policymakers, university authorities, national healthcare providers, and other relevant stakeholders to strengthen existing sexual health interventions and develop targeted policies that enhance awareness and understanding of sexually transmitted infections among KNUST students. It is essential to organize regular campaigns, health talks, and accessible screening and diagnostic sessions on campus to provide accurate information and early detection opportunities. These efforts should prioritize creating a supportive environment that encourages open dialogue and reduces stigma, ultimately empowering students to take proactive measures in protecting their reproductive health and wellbeing.

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APPENDIX 1

FIELD QUESTIONNAIRE

Awareness, Risk Perception and Preventive Measures for *SEXUALLY TRANSMITTED INFECTIONs* (STIs) Among Kwame Nkrumah University of Science and Technology Students

This survey is part of a research project titled "Awareness, Risk Perception, and Preventive Measures of STIs among Students at KNUST." Your participation is voluntary and your responses will be kept anonymous and confidential. Please answer the questions honestly.

SECTION 1- INFORMED CONSENT

This survey is part of a research project titled "Awareness, Risk Perception, and Preventive Measures of STIs among Students at KNUST." Your participation is voluntary and your responses will be kept anonymous and confidential. Please answer the questions honestly.

By continuing, you consent to participate in this study.

1. Do you consent to participate in this study? *

   Yes

   No

SECTION 2- DEMOGRAPHICS

2. **AGE** *

18	28	38
19	29	39
20	30	40
21	31	41
22	32	42
23	33	43
24	34	44
25	35	45
26	36	
27	37	

3. GENDER *

Male	Female	Prefer not to say

4. **LEVEL OF STUDY ***

100	300	500
200	400	600
POSTGRADUATE		

5. **COLLEGE ***

HEALTH SCIENCES	ARTS AND BUILT ENVIRONMENT
SCIENCE	ENGINEERING
HUMANITIES AND SOCIAL SCIENCES	AGRICULTURE AND NATURAL RESOURCES

6. **YOUR CURRENT RELATIONSHIP STATUS ***

Single	In a relationship	Married

7. **WHAT IS YOUR CURRENT SEXUAL ACTIVITY STATUS ***

In a committed relationship (not married)

In a committed relationship (married)

Engaging in casual sex (non-committed sexual encounters)

Not engaging in sexual activity

8. How many sexual partners have you had in the past 2 years *

0	3	6+
1	4	
2	5	

9. What is the total number of lifetime sexual partners? *

0	3	6
1	4	
2	5	

AWARENESS OF STIs

10. Have you heard of the following STIs? (Tick all that apply)

HIV/AIDS	Herpes
Chlamydia Gonorrhea	Hepatitis B
Syphyllis	Trichomoniasis
Human Papilloma Virus(HPV)	None of the above

11. Can STIs be transmitted without showing symptoms?

Yes | Maybe

No | I do not know

12. Which STI do you believe is most common among university students?

HIV/AIDS | Herpes

Gonohrrea | Hepatitis B

Chlamydia | Trichomoniasis

Syphyllis | Others

HPV

KNOWLEDGE ON STIs

MODE OF TRANSMISSION OF HIV/AIDS

13. Having unprotected sex

Yes No Maybe

14. Through mosquito Bites

Yes No Maybe

15. Through sharing of needles .

Yes No Maybe

16. Hugging someone with HIV/AIDS.

Yes No Maybe

17. Kissing

Yes No Maybe

MODE OF TRANSMISSION OF GONORRHEA

18. Unprotected vaginal, oral or anal sex.

Yes No Maybe

19. Using the same toilet seat

Yes No Maybe

20. From mother to baby during childbirth

Yes No Maybe

21. Wearing infected clothes

Yes No Maybe

MODE OF TRANSMISSION OF SYPHILLIS

22. Unprotected sexual contact (vaginal, oral or anal)

Yes No Maybe

23. Kissing someone with a syphilitic sore.

Yes No Maybe

24. Sharing toothbrushes or razors.

Yes No Maybe

25. From mother to fetus during pregnancy.

Yes No Maybe

26. Sitting on public toilets.

Yes No Maybe

MODE OF TRANSMISSION OF CHLAMYDIA

27. Unprotected vaginal, oral or anal sex.

Yes No Maybe

28. Sitting on public toilets.

Yes No Maybe

29. From mother to baby during childbirth.

Yes No Maybe

30. Touching infected clothing.

Yes No Maybe

31. Sharing food or drinks.

Yes No Maybe

SYMPTOMS

32. What is a common symptom of chlamydia?

Pain during urination Rash on hands

Coughing I do not know

33. What is a common symptom of gonorrhea?

Vaginal/penile discharge Fever

Vomiting Headache

I do not know

34. What is a common symptom of syphilis?

Painless sore (chancre) | Stomach pain

Itchy eyes | I do not know

35. What is a common symptom of HIV??

Fatigue | Weight loss

Night sweats | I do not know

SOURCES OF INFORMATION

36. Where do you primarily get information about STIs? (Tick all that apply)

Social media | Parents/family members

Friends or peers | Healthcare professionals

Television/radio- Health talks | School/university courses

Workshops/Seminars | Newspapers or magazines

Internet sources | I have never received STI information

37. Which of these sources do you trust the most for accurate information on STIs? (Tick all that apply)

Social media | Parents/family members

Friends or peers | Healthcare professionals

Television/radio- Health talks | School/university courses

Workshops/Seminars | Newspapers or magazines

Internet sources | I have never received STI information

38. Have you ever attended any STI-related education session or workshop on campus?

Yes | No | I do not remember

RISK PERCEPTION

39. Do you think STIs can be transmitted even if the infected person has no symptoms?

Yes | No | Maybe

40. Is it true that some people with gonorrhoea do not show any symptoms?

Yes | Maybe

No | I do not know

41. Is it true that some people with HIV do not show any symptoms?

Yes Maybe

No I do not know

42. Is it true that some people with Syphilis do not show any symptoms?

Yes Maybe

No I do not know

43. Is it true that some people with Chlamydia do not show any symptoms?

Yes Maybe

No I do not known

44. Gonorrhoea can spread to other organs beyond the genitals

Yes Maybe

No I do not know

45. The effects of syphilis go away completely once the rash disappears

Yes Maybe

No I do not know

46. HIV can cause long-term health problems even before it progresses to AIDS

Yes Maybe

No I do not know

47. Can untreated chlamydia cause infertility in women or men?

Yes Maybe

No I do not know

48. How likely is it that students at KNUST are at risk of contracting STIs?

1 2 3 4 5

Less likely Very Likely

49. Do you consider yourself at risk of contracting an STI?

Yes No Maybe

50. How concerned are you about STIs?

1 2 3 4 5

Less Concerned Very Concerned

51. Do you believe having multiple sexual partners increases the risk of STIs?

Yes No Maybe

52. Do you think condom use is necessary with a trusted partner?

Yes No Maybe

PREVENTIVE BEHAVIOURS

53. Do you consistently use condoms during sexual intercourse?

Always | Never
Sometimes | I'm not sexually active

54. Have you ever been tested for any STI?

Yes | No

55. Have you ever tested positive for any STI?

Yes | No

56. Are you aware of STI testing services available on or near campus?

Yes | No

57. Would you inform a sexual partner if you were diagnosed with an STI?

Yes | No | Maybe

58. In your opinion, what is the most effective way to prevent STIs among university students?